

SYLLABUS TOPICS COVERED

- Java Buzzwords
- Data Types
- Variables & Operators
- Control Structures
- Arrays
- Console I/O
- Command-Line Arguments

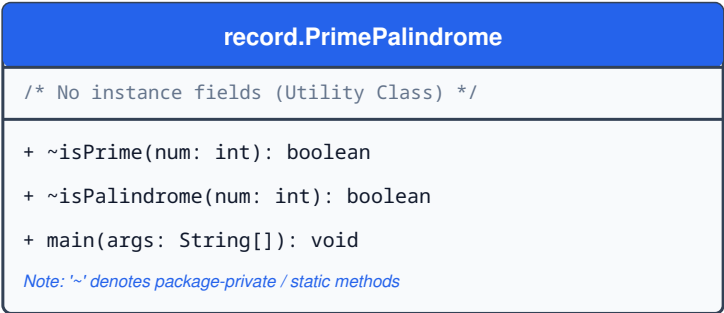
PROGRAM 1: PRIME PALINDROME CHECKER

Problem Statement [CO1]: Write a sample Java program to check whether a given integer number is a prime palindrome or not.

Step-by-Step Process & Algorithm:

- Input Reading:** Initialize a `Scanner` object to read an integer from standard console input (`System.in`).
- Prime Verification (`isPrime`):** Check if number `$num` ≤ 1 (not prime). Otherwise, iterate `$i` from `$2` up to `$\lfloor \sqrt{\text{num}} \rfloor`. If `$num % i == 0`, return `false`. If no divisor is found, return `true`.
- Palindrome Verification (`isPalindrome`):** Store initial number in `original`. Reverse digits using `$digit = num % 10` and `$reversed = (reversed * 10) + digit` in a loop while `$num > 0`. Return `true` if `$original == reversed`.
- Combined Logic:** Evaluate `isPrime(number) && isPalindrome(number)`. If both evaluate to `true`, print that the number is a Prime Palindrome; otherwise, output that it is NOT.
- Resource Cleanup:** Close the `Scanner` stream to prevent resource leaks.

UML CLASS DIAGRAM — PRIMEPALINDROME



Java Source Code:

```
package record;

import java.util.Scanner;

public class PrimePalindrome {

    // checking whether a number is a prime or not
```

```

static boolean isPrime(int num) {
    if (num <= 1)
        return false;
    for (int i = 2; i <= Math.sqrt(num); i++) {
        if (num % i == 0)
            return false;
    }
    return true;
}

// checking whether a number is a palindrome or not
static boolean isPalindrome(int num) {
    int original = num;
    int reversed = 0;

    while (num > 0) {
        int digit = num % 10;
        reversed = (reversed * 10) + digit;
        num /= 10;
    }

    return original == reversed;
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter a number: ");
    int number = scanner.nextInt();

    if (isPrime(number) && isPalindrome(number)) {
        System.out.println(number + " is a Prime Palindrome.");
    } else {
        System.out.println(number + " is NOT a Prime Palindrome.");
    }

    scanner.close();
}

```

SAMPLE CONSOLE OUTPUT (TEST CASE 1)

```

Enter a number: 131
131 is a Prime Palindrome.

```

SAMPLE CONSOLE OUTPUT (TEST CASE 2)

```

Enter a number: 120
120 is NOT a Prime Palindrome.

```

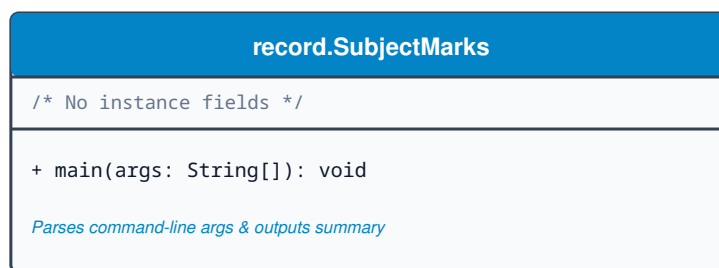
PROGRAM 2: COMMAND-LINE SUBJECT MARKS PROCESSOR

Problem Statement: Write a Java program which reads marks of five subjects through command line arguments and prints the total and average marks.

Step-by-Step Process & Algorithm:

1. **Argument Validation:** Check if `args.length < 5`. If true, display an error message explaining usage and terminate execution early using `return`.
2. **Array Allocation:** Allocate a `double[]` array of size 5 to store numerical values of marks.
3. **Parsing & Accumulation:** Iterate through command-line arguments using a `for` loop (`$i = 0 \dots 4$`). Parse each string argument into `double` using `Double.parseDouble(args[i])` and add it to `total`.
4. **Average Calculation:** Compute `$average = total / 5.0$`.
5. **Formatted Output Display:** Print individual subject marks, total marks, and average formatted to two decimal places using `System.out.printf()`.

UML CLASS DIAGRAM — SUBJECTMARKS



Java Source Code:

```
package record;

public class SubjectMarks {

    public static void main(String[] args) {
        if (args.length < 5) {

            System.out.println("Error: Please provide marks for 5 subjects as command-line arguments.");
            System.out.println("Usage: java SubjectMarks <mark1> <mark2> <mark3> <mark4> <mark5>");
            return;
        }

        double[] marks = new double[5];
        double total = 0;

        for (int i = 0; i < 5; i++) {
            marks[i] = Double.parseDouble(args[i]);
            total += marks[i];
        }

        double average = total / 5.0;

        // Output results
        System.out.println("--- Student Marks Summary ---");
        for (int i = 0; i < 5; i++) {
            System.out.printf("Subject %d: %s\n", i + 1, marks[i]);
        }
    }
}
```

```
        System.out.println("-----");
        System.out.println("Total Marks   : " + total);
        System.out.printf("Average Marks : %.2f
", average);
    }
}
```

SAMPLE EXECUTION & OUTPUT

```
$ java record.SubjectMarks 85.5 90.0 78.25 92 88.5
```

```
--- Student Marks Summary ---
```

```
Subject 1: 85.5
```

```
Subject 2: 90.0
```

```
Subject 3: 78.25
```

```
Subject 4: 92.0
```

```
Subject 5: 88.5
```

```
-----
```

```
Total Marks : 434.25
```

```
Average Marks : 86.85
```